© 2020 – Author and Collision Publishing

Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

Robert Anderson
Biomechanics Analysis

Wesley Vandiver
Collision Forensics, Inc.

EDR SUMMIT
Research • Collection • Analysis

1

Robert D Anderson

- Biomechanical Engineer/Biomechanics Analysis - collision reconstruction and injury mechanics
- ARC-CSI crash instrumentation back to 2006
- SATAI's crash program since 2007
- Developed Phantom Driver with Michael Rosenfield
- Dozens of crash related publications
- Co-founded crash·ol·o·gy, The Science of Crashes, an ongoing research project that is a recurring feature in Collision Magazine, designed to educate collision investigators on current topics, new developments, and crash research



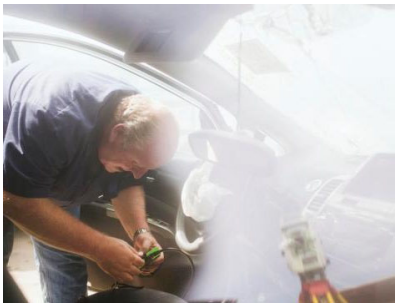
(c) 2020 EDR Users S

2

© 2020 – Author and Collision Publishing

Wesley Vandiver

- Director of Training at Berla Corporation
- President of Collision Forensics, Inc.
- Collision analysis background:
  - 2-years Senior Consultant with a forensic engineering
  - 28-year law enforcement career
- Multiple crash related publications
- Co-founded crash·ol·o·gy, The Science of Crashes, an ongoing research project that is a recurring feature in Collision Magazine, designed to educate collision investigators on current topics, new developments, and crash research



(c) 2020 EDR Users Summit

3

Introduction

Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

4

© 2020 – Author and Collision Publishing

Introduction



Unlike traditional Event Data Recorders (EDRs), which have become a vital source of information for those analyzing collision, the Toyota Vehicle Control History (VCH) can include data from a number of events/triggers that are not collision related.

(c) 2020 EDR Users Summit

Acquiring Toyota VCH data requires different procedures than acquiring traditional EDR data, but yields some familiar data elements, including vehicle speed, engine RPM, accelerator pedal percentage, etc.

5

Introduction



First introduced to Accident Reconstruction community by Dave Hallman at the 2014 CDR User's Summit.

Recording by Non-Crash Trigger - Iyoda (TMC) in SAE 2016-01-1495

Accuracy of VCH during AEB – Xing (MEA) in SAE 2018-01-1441

(c) 2020 EDR Users Summit

6

© 2020 – Author and Collision Publishing

Introduction



In this project a series of instrumented tests were conducted to examine the potential non-AEB data available from the Toyota VCH and the accuracy of that data.

Additionally, some best practices became apparent that could assist an investigator in explaining recovered data.

(c) 2020 EDR Users Summit

7

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020



Outline

Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

8

© 2020 – Author and Collision Publishing

Outline



- Mechanics of Techstream
  - Software installation, reactivation and obtaining Data
- Vehicle Control History (VCH) Triggers
- Testing and Instrumentation
- Verification of Speed and Acceleration Data
- Ignition Cycles
- Time Stamps

(c) 2020 EDR Users Summit

9

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020



Installing TechStream Software

Testing and Validation of Toyota Vehicle Control History Data

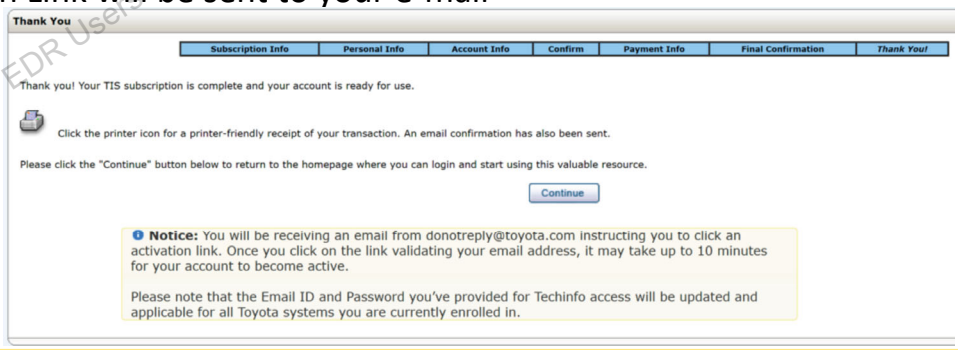
(c) 2020 EDR Users Summit

10

© 2020 – Author and Collision Publishing

TechStream Software Install and Setup

- https://techinfo.snapon.com/TIS/register.aspx
- Continue to purchase 2 day subscription of Professional Diagnostic ($65)
- An Activation Link will be sent to your e-mail



(c) 2020 EDR Users Summit

March 9-11, 2020 Slide 11

11

TechStream Software Install and Setup

- May take up to 10 minutes for account to become active
- TIS Subscriber Login
- TIS Tab, Diagnostics, ScanTool – Techstream Software , Full Install

(c) 2020 EDR Users Summit

March 9-11, 2020 Slide 12

12

© 2020 – Author and Collision Publishing



13

TechStream Software Install and Setup

- Run the install file as administrator
- Open 

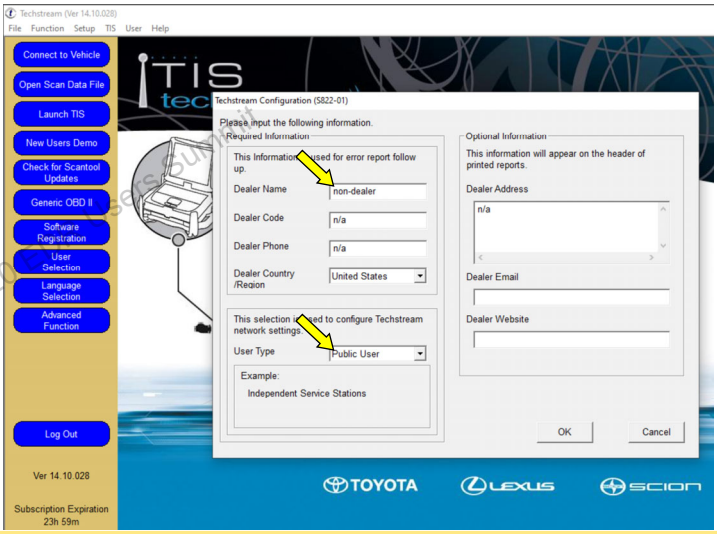
- Configure Techstream, set user type to “Public User”
- Under Setup, select VIM Select (CDR900, Mongoose, etc.)

(c) 2020 EDR Users Summit

14

© 2020 – Author and Collision Publishing

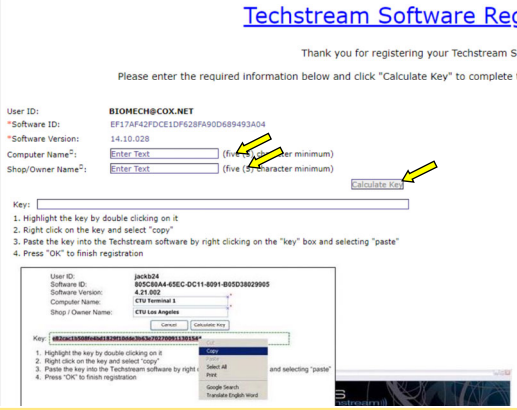
TechStream Software Install and Setup



15

TechStream Software Install and Setup

- Click on Software Registration, then again on Software Registration and follow instructions to create a key



16

© 2020 – Author and Collision Publishing

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020

crash·ol·o·gy
THE SCIENCE OF CRASHES

Extending (reactivating) My Subscritpion

(c) 2020 EDR Users Summit

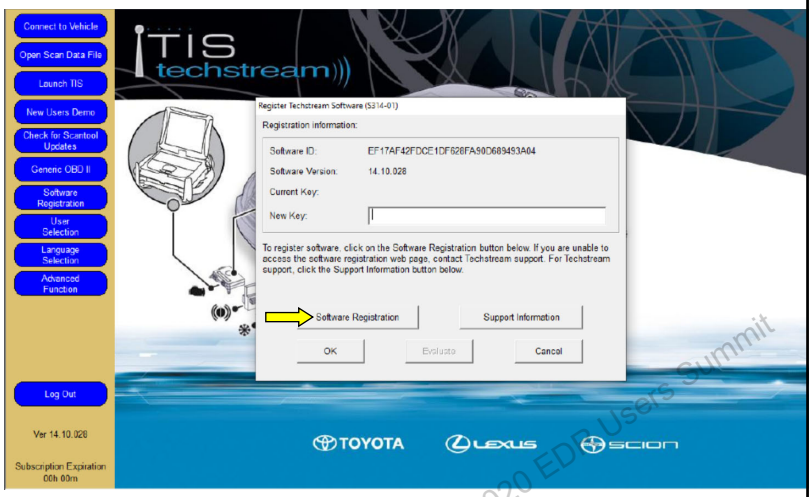
Testing and Validation of Toyota Vehicle Control History Data

17

TechStream Extend My Subscription

crash·ol·o·gy
THE SCIENCE OF CRASHES

- Open Techstream
- Click Software Registration



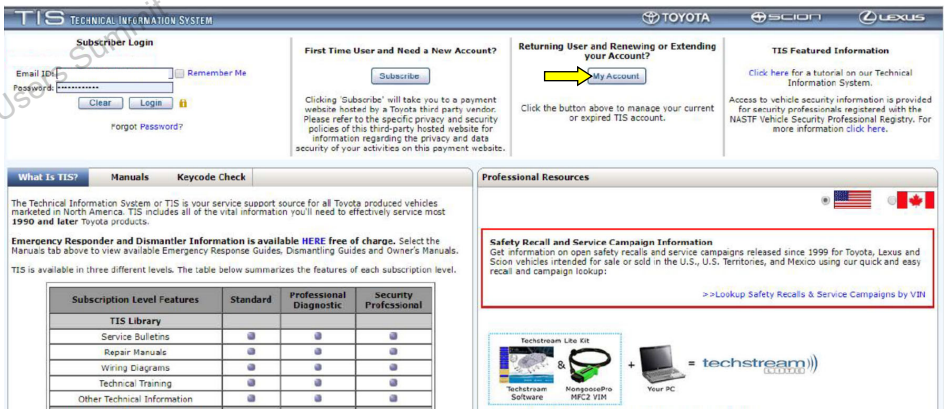
(c) 2020 EDR Users Summit

18

© 2020 – Author and Collision Publishing

TechStream Extend My Subscription

- Select My Account

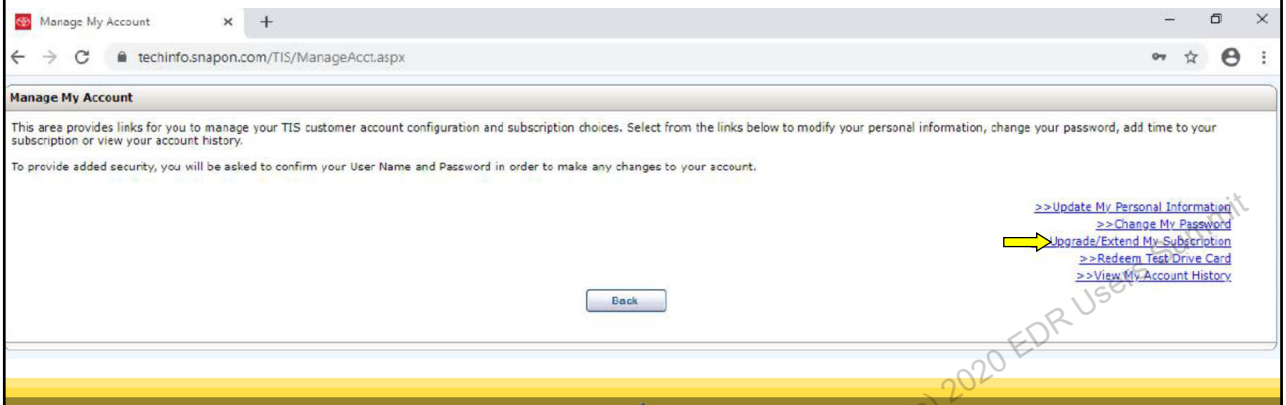


(c) 2020 EDR Users Summit

19

TechStream Extend My Subscription

- Upgrade/Extend My Subscription



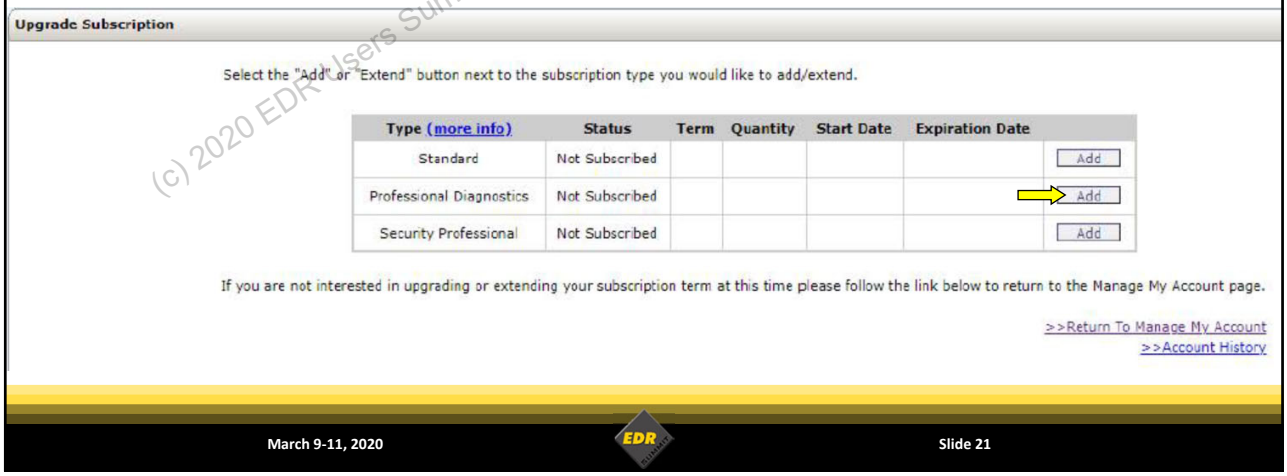
(c) 2020 EDR Users Summit

20

© 2020 – Author and Collision Publishing

TechStream Extend My Subscription

- Account Maintenance Login
- Add Professional Diagnostics Subscription

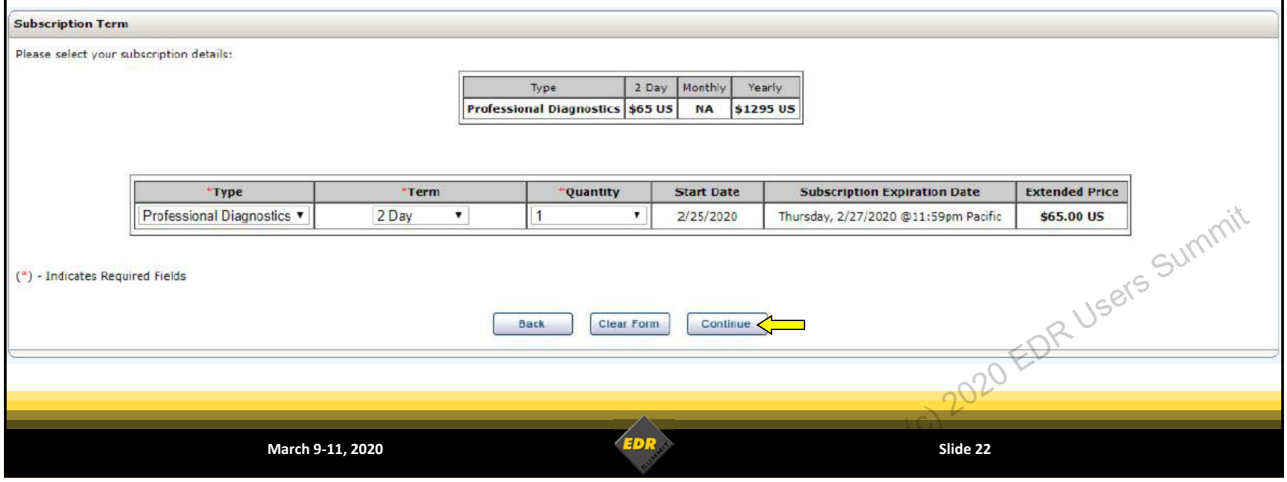


(c) 2020 EDR Users Summit

21

TechStream Extend My Subscription

- Subscription Term – 2 days ($65) or 1 year ($1295)
- Continue and pay with credit card



(c) 2020 EDR Users Summit

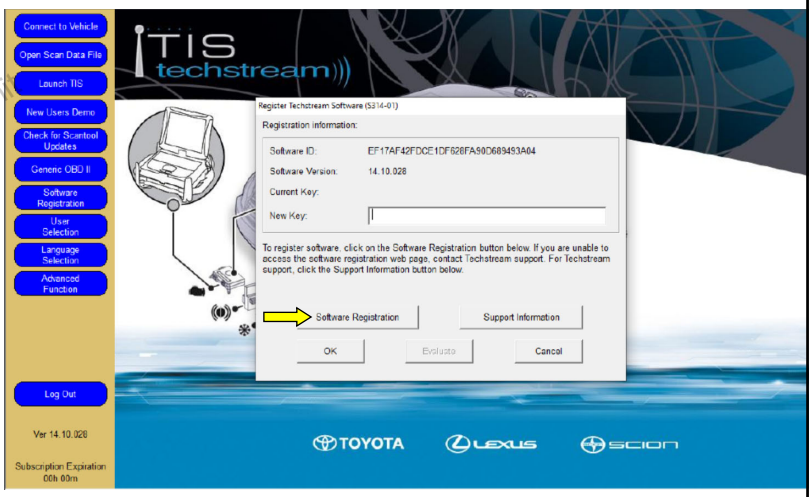
22

© 2020 – Author and Collision Publishing

TechStream Extend My Subscription

- Open 

- Click Software Registration

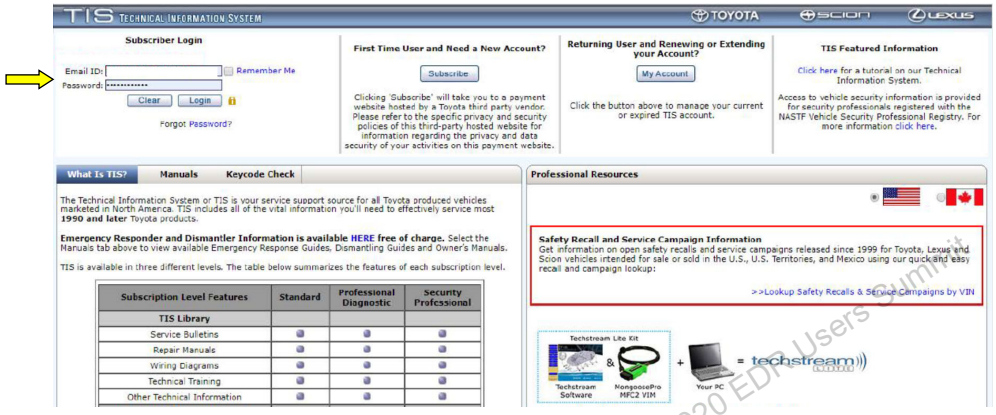


(c) 2020 EDR Users Summit

23

TechStream Extend My Subscription

- Wait up to 10 minutes and then Subscriber Login

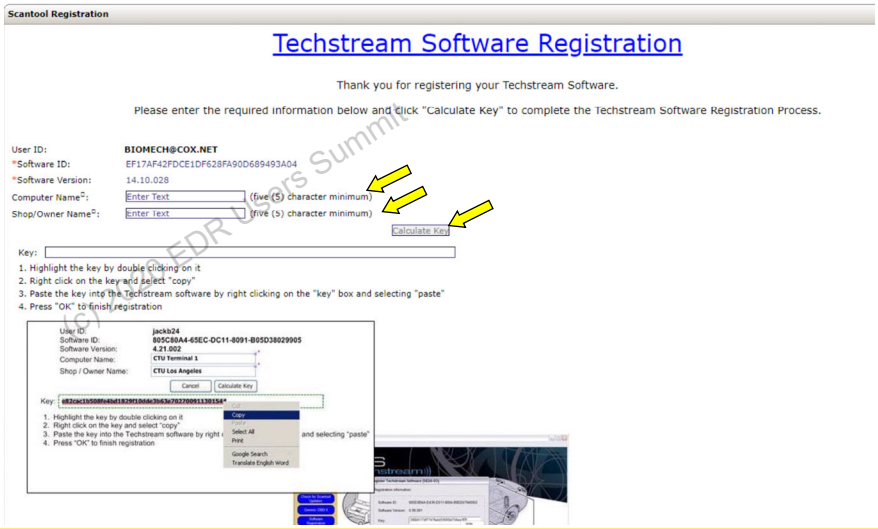


(c) 2020 EDR Users Summit

24

© 2020 – Author and Collision Publishing

TechStream Extend My Subscription



(c) 2020 EDR Users Summit

25

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020

Obtaining TechStream/Scantool Data

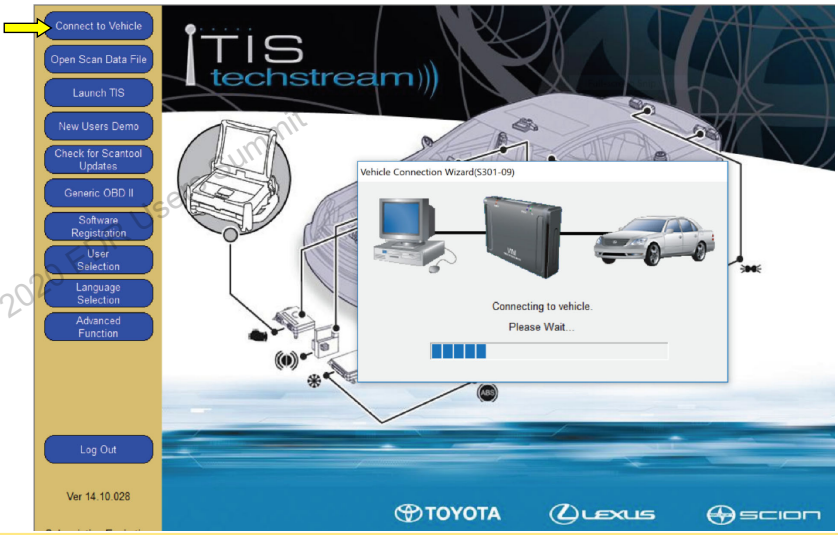
Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

26

© 2020 – Author and Collision Publishing

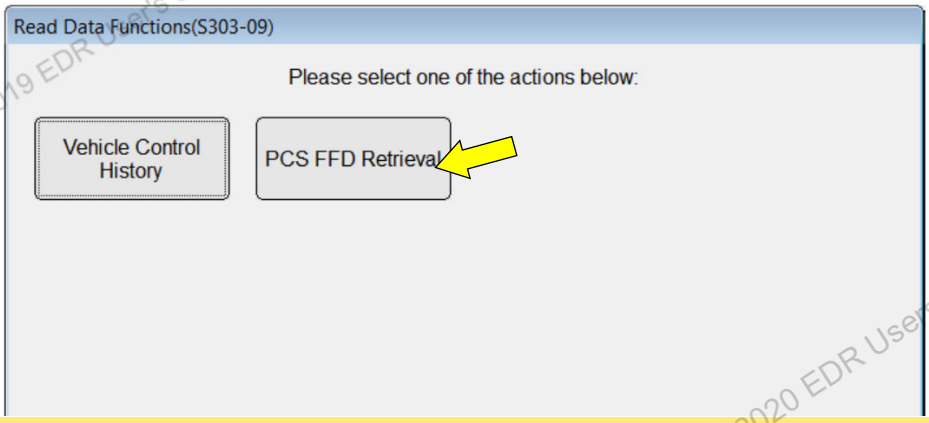
TechStream Connect to Vehicle



March 9-11, 2020 Slide 27

27

TechStream Read Data

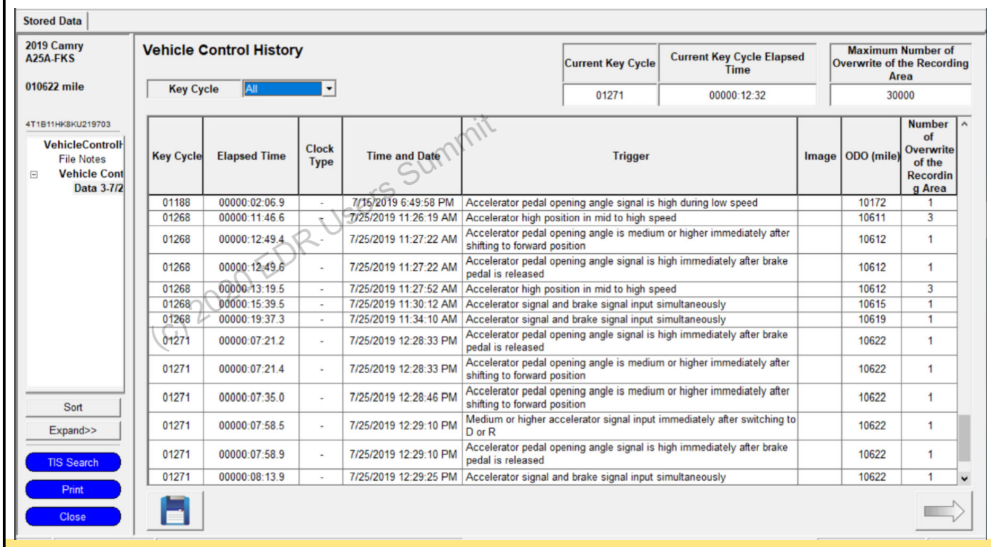


March 9-11, 2020 Slide 28

28

© 2020 – Author and Collision Publishing

TechStream Stored Data



29

TechStream Save the VCH report

Vehicle Control History Report
(1 of 3)

Current Key Cycle : 01271
Current Key Cycle Elapsed Time : 00000:12:32
Maximum Number of Overwrite of the Recording Area : 30000
Key Cycle : All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area
00013	00000:00:15.5	-	1/31/2019 3:19:39 PM	Medium or higher accelerator signal input immediately after switching to D or R		0	1
00020	00000:00:54.8	-	2/1/2019 6:38:54 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to R		1	1
00113	00000:23:58.6	-	2/25/2019 3:10:58 PM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		760	1
00129	00000:05:11.2	-	2/27/2019 7:23:31 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		846	1
00130	00000:02:13.9	-	2/27/2019 7:45:17 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		846	1
00143	00001:03:54.7	-	2/28/2019 7:42:36 PM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		959	1
00144	00000:01:29.7	-	2/28/2019 8:20:15 PM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		980	1
00152	00000:00:53.2	-	3/1/2019 10:34:24 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		1001	1
00161	00000:05:45.1	-	3/3/2019 1:25:18 AM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		1014	1
00164	00000:13:37.4	-	3/3/2019 3:03:50 PM	Accelerator pedal opening angle signal is high during low speed		1026	1
00190	00000:25:26.7	-	3/7/2019 9:40:49 AM	Medium or higher accelerator signal input immediately after switching to D or R		1084	1
00191	00000:00:46.4	-	3/7/2019 10:20:31 AM	Accelerator signal and brake signal input simultaneously		1085	1
00191	00000:02:30.3	-	3/7/2019 10:22:15 AM	Accelerator signal and brake signal input simultaneously		1085	1
00212	00000:02:49.7	-	3/13/2019 2:55:20 PM	Medium or higher accelerator signal input immediately after switching to D or R		1469	1
00212	00000:02:49.8	-	3/13/2019 2:55:20 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		1469	1
00213	00000:06:46.1	-	3/13/2019 3:10:26 PM	Medium or higher accelerator signal input immediately after switching to D or R		1469	1
00233	00000:36:54.3	-	3/14/2019 8:35:29 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		1557	1

30

© 2020 – Author and Collision Publishing

TechStream VCH CSV

	A	B	C	D	E	F	G	H	I
3	Time	Data ID	Trigger ID	Trigger ID (Sub)	Trigger Time Information	Detection Counts of Certain Level of Acceleration	Detection Counts of Certain Level of Acceleration	Number of Overwrite of the Recording Area	Time and Date
4	sec				ms	ms			
4114	623.35	N14	4						
4115	623.85	N15	4						
4116									
4117	611.35	M0		5					
4118	611.85	M1		5					
4119	612.35	M2		5					
4120	612.85	M3		5					
4121	613.35	M4		5					
4122	613.85	M5		5					
4123	614.35	M6		5					
4124	614.85	M7		5					
4125	615.35	M8		5					
4126	615.85	M9		5					
4127	616	T		5	2	350	32766.5	65533	2 7/25/2019 15:54
4128	616.35	N0		5					
4129	616.85	N1		5					
4130	617.35	N2		5					
4131	617.85	N3		5					
4132	618.35	N4		5					
4133	618.85	N5		5					
4134	619.35	N6		5					
4135	619.85	N7		5					
4136	620.35	N8		5					
4137	620.85	N9		5					
4138	621.35	N10		5					
4139									
4140									

(c) 2020 EDR Users Summit

31

TechStream VCH CSV

	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE
3	Successive Trigger Time Information	Vehicle Speed	Accelerator Opening Ratio	Engine RPM Data	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure	Yaw Rate Sensor Signal	Longitudinal G Sensor Signal	Lateral G Sensor Signal	Throttle Opening Ratio	Command Value of Fuel Injection Amount	Steering Signal
4	ms	km/h	%	rpm				MPa	deg/s	m/s^2	m/s^2	%	mm3/st	deg
4114		55	100	4200	0	1	0	0	-0.488	1.65094	-0.14356	100	100	0
4115		59	100	4500	0	1	0	0	-0.488	2.08162	-0.07178	99.5	99.5	0
4116														
4117		0	0	700	0	3	1	0.816	-0.488	0	-0.07178	0	0	22.5
4118		0	0	700	0	3	1	0.816	-0.488	0	-0.07178	0	0	22.5
4119		0	0	700	0	3	1	0.816	-0.488	0.07178	-0.07178	0	0	22.5
4120		0	0	700	0	3	0	0	-0.488	0	-0.07178	0	0	22.5
4121		0	0	700	0	3	0	0	-0.488	0	-0.07178	0	0	22.5
4122		0	0	700	0	3	0	0	-0.488	0.07178	-0.07178	0	0	22.5
4123		0	0	700	0	3	0	0	-0.488	0	-0.07178	0	0	22.5
4124		0	0	700	0	3	1	2.064	-0.488	0.07178	-0.07178	0	0	22.5
4125		0	0	700	0	2	1	2.208	-0.488	0	-0.07178	0	0	22.5
4126		0	0	700	0	1	0	0	-0.488	0	0	0	0	22.5
4127		0	88	800	0	1	0	0	-0.488	0.07178	-0.07178	19.5	19.5	22.5
4128		0	100	1000	0	1	0	0	-0.488	2.00984	-0.07178	12.5	12.5	22.5
4129		3	100	1300	0	1	0	0	0	2.08162	-0.07178	100	100	27
4130		7	100	1500	0	1	0	0	0.488	2.1534	0	99.5	99.5	22.5
4131		10	100	1600	0	1	0	0	0.976	1.65094	0	100	100	18
4132		13	100	1900	0	1	0	0	0	1.86628	-0.28712	99.5	99.5	0
4133		16	100	2300	0	1	0	0	-0.488	2.1534	-0.14356	99.5	99.5	0
4134		20	100	2800	0	1	0	0	-0.976	2.58408	-0.14356	99.5	99.5	-4.5
4135		24	100	3300	0	1	0	0	-0.976	2.5123	0	99.5	99.5	0
4136		29	100	3900	0	1	0	0	-0.488	2.79942	-0.14356	99.5	99.5	0
4137		33	100	4500	0	1	0	0	-0.488	2.72764	-0.14356	99.5	99.5	0
4138		38	100	5100	0	1	0	0	-0.976	3.01476	-0.21534	100	100	0
4139														

(c) 2020 EDR Users Summit

32

© 2020 – Author and Collision Publishing

TechStream VCH CSV



IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	JJ	JK
FHL3 Object Distance X m	FHL3 Relative Speed Y m/s	FHL3 Relative Speed X m/s	FHL3 RCS Level	FHL3 TTC sec	FHL3 (New Target Flag)	FHL3 (Detecting Radar)	FHL3 (Operating Trigger Flag)	FHL3 (Ghosting Flag)	FHL3 (Extrapolating Flag)	FHL3 (Radar Blockage Ratio Flag)	FHL3 (Radar Distortional Flag)	FHL3 (Adjacent Reflection Flag)	FHL3 (CAN Received Data)	FHL3 ID

271 Columns!

(c) 2020 EDR Users Summit

33

TechStream VCH Example



Vehicle Control History Report
(1 of 2)

Current Key Cycle : 04404
Current Key Cycle Elapsed Time : 00000:22:37
Maximum Number of Overwrite of the Recording Area : 30000

Key Cycle : All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (mile)	Number of Overwrite of the Recording Area
00078	00000:05:35.6	-	-	Accelerator signal and brake signal input simultaneously	258	1
⋮				⋮		⋮
01113	00000:09:13.2	-	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7132	1
01168	00000:02:20.2	-	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	7321	1
01220	00000:12:51.9	-	-	Accelerator signal and brake signal input simultaneously	7502	1
01281	00000:37:27.1	-	-	Accelerator signal and brake signal input simultaneously	7726	1
01365	00000:02:15.0	-	-	Sudden braking history	8053	1
01409	00000:06:23.2	-	-	Accelerator signal and brake signal input simultaneously	8204	1
01476	00000:34:07.8	-	-	Accelerator signal and brake signal input simultaneously	8542	1
01516	00000:05:03.8	-	-	Accelerator signal and brake signal input simultaneously	8644	1

(c) 2020 EDR Users Summit

34

© 2020 – Author and Collision Publishing

TechStream VCH Example



(c) 2020 EDR Users Summit

35

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020

VCH Basics & Events (“Triggers”)

Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

36

© 2020 – Author and Collision Publishing

VCH Basics

- The VCH is a function that records control data (record data) when triggered by specific vehicle behavior
- The VCH can also store image data when a specific trigger is set
- The VCH is recorded in different storage areas designated by trigger group and it is possible to save up to 140 items in total
- If storage space is filled, the data is overwritten on a First-In, First-Out (FIFO) basis
- Each trigger receives time information from the main body Electronic Control Unit (ECU), which is the time elapsed since the engine switch was turned on for the given ignition cycle
- Absolute time may be recorded for vehicles with factory-installed navigation systems.

(c) 2020 EDR Users Summit

37

Examples of Triggers

2018-2019 MY Toyota Camry

Trigger Item ID No.	Trigger Item Item	Record Data Before Detection Recoding Time (sec.)	Before Detection Number of Points	After Detection Recording Time (sec.)	After Detection Number of Points	Frequency (sec.)
4	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7.5	15	7.5	16	0.5
5-1	Accelerator pedal opening angle is medium or higher immediately after shifting to R	5	10	5	11	0.5
5-2	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position					
5-3	Accelerator pedal opening angle signal is high immediately after brake pedal is released					
8	Accelerator signal and brake signal input simultaneously	5	10	5	11	0.5
9	Medium or higher accelerator signal input immediately after switching to D or R	5	10	5	11	0.5
10	Medium or higher accelerator signal input during N	5	10	5	11	0.5
12	VSC operation history	5	34	5	35	0.15
13	TRC operation history	5	34	5	35	0.15
14	ABS operation history	5	34	5	35	0.15
15	Sudden braking history	5	34	5	35	0.15
16	Sudden turning history	5	34	5	35	0.15
17	Accelerator pedal opening angle signal is high during cruise control operation	5	10	5	11	0.5
18-1	PCS operation history (warning buzzer operation)	5	10	4	9	0.5
18-2	PCS operation history (warning brake operation)					
18-3	PCS operation history (pre-collision brake assist operation)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
18-15	PCS operation history (side airbag operation request)					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	8	20	0	1	0.4

(c) 2020 EDR Users Summit

38

© 2020 – Author and Collision Publishing

Examples of Triggers

01268	00000:11:46.6	-	7/25/2019 11:26:19 AM	Accelerator high position in mid to high speed		10611	3
01268	00000:12:49.4	-	7/25/2019 11:27:22 AM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		10612	1
01268	00000:12:49.6	-	7/25/2019 11:27:22 AM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10612	1
01268	00000:13:19.5	-	7/25/2019 11:27:52 AM	Accelerator high position in mid to high speed		10612	3
01268	00000:15:39.5	-	7/25/2019 11:30:12 AM	Accelerator signal and brake signal input simultaneously		10615	1
01268	00000:19:37.3	-	7/25/2019 11:34:10 AM	Accelerator signal and brake signal input simultaneously		10619	1
01271	00000:07:21.2	-	7/25/2019 12:28:33 PM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10622	1
01271	00000:07:21.4	-	7/25/2019 12:28:33 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		10622	1
01271	00000:07:35.0	-	7/25/2019 12:28:46 PM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		10622	1
01271	00000:07:58.5	-	7/25/2019 12:29:10 PM	Medium or higher accelerator signal input immediately after switching to D or R		10622	1
01271	00000:07:58.9	-	7/25/2019 12:29:10 PM	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10622	1
01271	00000:08:13.9	-	7/25/2019 12:29:25 PM	Accelerator signal and brake signal input simultaneously		10622	1
01271	00000:08:14.1	-	7/25/2019 12:29:26 PM	ABS operation history		10622	4
01271	00000:08:14.2	-	7/25/2019 12:29:26 PM	Sudden braking history		10622	1
01271	00000:08:14.4	-	7/25/2019 12:29:26 PM	ABS operation history		10622	4
01271	00000:08:14.4	-	7/25/2019 12:29:26 PM	Accelerator high position in mid to high speed		10622	3

(c) 2020 EDR Users Summit

39

Data Elements

Record Data Chart

ITEM NAME
Vehicle Speed
Accelerator Opening Ratio
Engine RPM Data
Cruise Control
Shift Position Signal
Brake SW
Brake Oil Pressure
Yaw Rate Sensor Signal
Longitudinal G Sensor Signal
Lateral G Sensor Signal
Throttle Opening Ratio
Command Value of Fuel Injection Amount

Steering Signal
HV Recognition Flag
Diesel Flag
Sensor Type Identification (Security)
Sensor Type Identification (G Sensor Layout)
Sensor Type Identification (G Sensor)
Sensor Type Identification (Yaw Rate Sensor)
Yaw G Sensor Installation Information
Shift Gear
PWR Mode
ECO Mode
Integrated Sport Mode Indicator
SNOW Mode
Valid/Invalid of Driving Mode
VSC OFF Lamp
TRC OFF Lamp
Valid/Invalid of TRC/VSC OFF Lamp

EV Mode
Valid/Invalid of EV Mode
Integrated Sport Mode Signal
Engine Load Factor
READY Signal
Gear Position Signal
Turbine RPM/Input Shaft RPM
Under VSC Control
Under ABS Control
Under VDM Control
ITEM NAME
Under TRC Operation Flag
FR Wheel Speed
FL Wheel Speed
RR Wheel Speed
RL Wheel Speed
Trigger Judgment Information
Driving Force Demand Judgment Flag
DSS Judgment Flag
Cruise Control Res/Accel SW
Cruise Control Set/Coast SW
Cruise Control Cancel SW
Cruise Control Brake SW

Cruise Control D Signal
Cruise Control Main SW
Cruise Control Set Speed
Vehicle Speed (during Cruise Control)
Cruise Control Request Value
ACC Target Distance
ACC Target Condition
ACC Relative Speed
PCS Reception Data Invalid Flag
ALM Request Flag
ABK Request Flag
PBA Request Flag
FPB Request Flag
PB Request Flag
Extension Request Flag
PSB Request Flag
PCS READY Flag
PCS Display (PSB Communication)
PCS Display (System Fault)
PCS Display (Radar Dirt)
PCS Display (Temporarily not available)
Target Distance
Target Relative Velocity

Target Lateral Position
Target Number
Detection of Certain Level of Acceleration
LKA DDR Trigger Signal
Invalid of LKA DDR Data
Steering Wheel Release Condition
Steering Operation of LKA
Steering Operation of LDA
LKA Control Condition (EPS Control)
LKA Control Condition
Turn Signal Operation
EPS Torque Sensor Value

(c) 2020 EDR Users Summit

40

© 2020 – Author and Collision Publishing

Testing Instrumentation and Equipment

(c) 2020 EDR Users Summit

Testing and Validation of Toyota Vehicle Control History Data

41

Test Vehicle, 2019 Toyota Camry

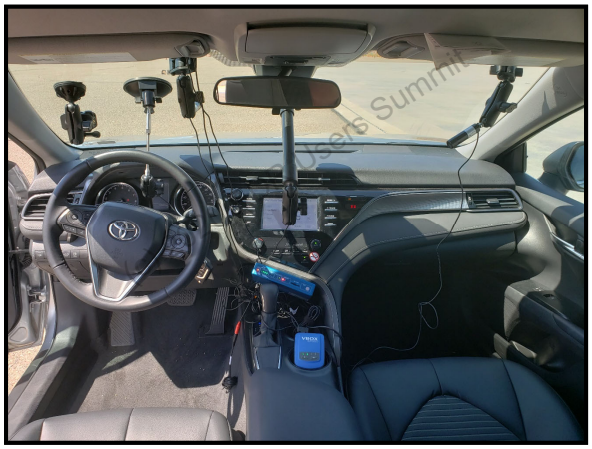


(c) 2020 EDR Users Summit

42

© 2020 – Author and Collision Publishing

Instrumentation & Equipment

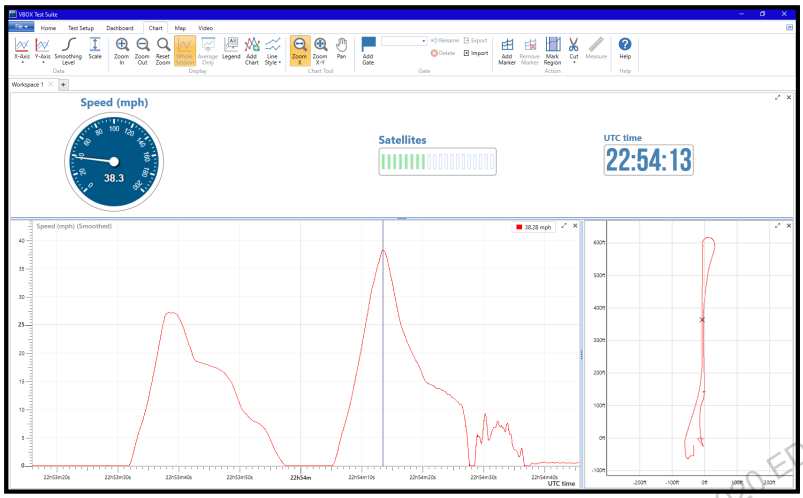


- Video VBOX Pro
- VBOX Sport
- Garmin VIRB
- Bosch CDR 900
- Toyota Techstream software
- Bosch CAN*plus* Vehicle Interface Module

March 9-11, 2020 Slide 43

43

VBOX Sport



March 9-11, 2020 Slide 44

44

© 2020 – Author and Collision Publishing

Bosch CDR 900 & Toyota Techstream

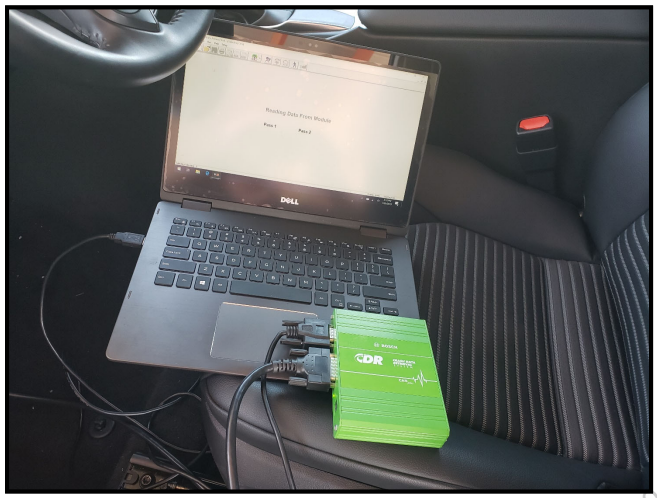


(c) 2020 EDR

March 9-11, 2020 Slide 45

45

Bosch CAN*plus* Vehicle Interface Module



(c) 2020 EDR Users Summit

March 9-11, 2020 Slide 46

46

© 2020 – Author and Collision Publishing

Verification of Speed and Acceleration Data

(c) 2020 EDR Users Summit

Testing and Validation of Toyota Vehicle Control History Data

47

Instrumentation: Racelogic VBOX

- Measures speed using Doppler shift of GPS radio signals
- Velocity
  - Accuracy: 0.1 kph (0.06 mph) when averaged over 4 samples
  - Resolution: 0.01 kph (0.006 mph)
- Acceleration
  - Accuracy: 0.5 %
  - Resolution: 0.01 G

(c) 2020 EDR Users Summit

48

© 2020 – Author and Collision Publishing

Instrumentation: Racelogic VBOX

- VBOX Sport
  - Speed
  - Lateral and longitudinal acceleration
  - Internal battery and antennae
  - No video
  - 20 Hz
- Video VBOX Pro
  - Speed
  - Lateral and longitudinal acceleration
  - External battery and antennae
  - Up to 4 videos
  - 10 Hz

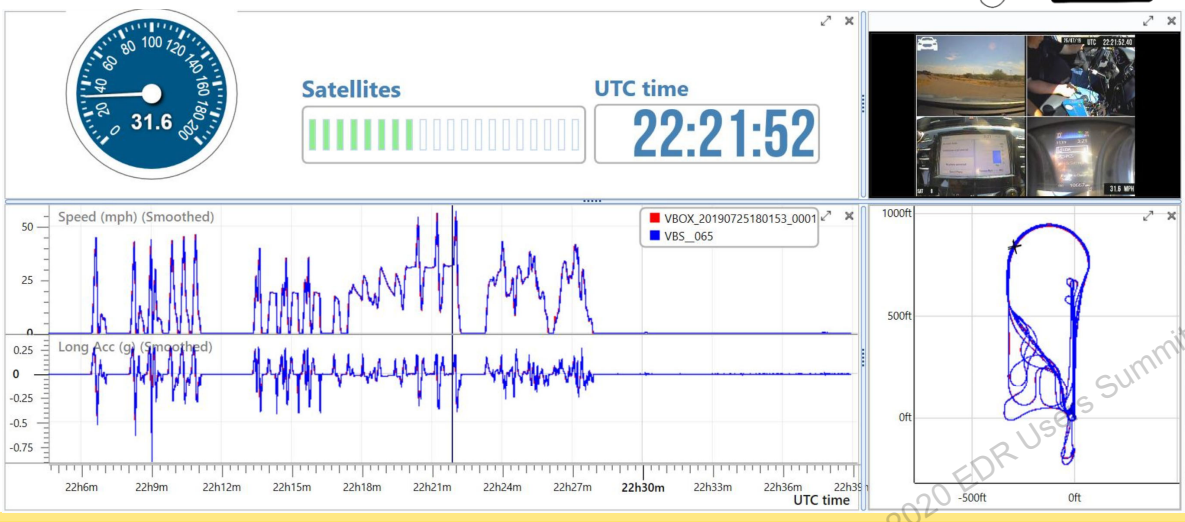
(c) 2020 EDR Users Summit





49

Instrumentation: Racelogic VBOX

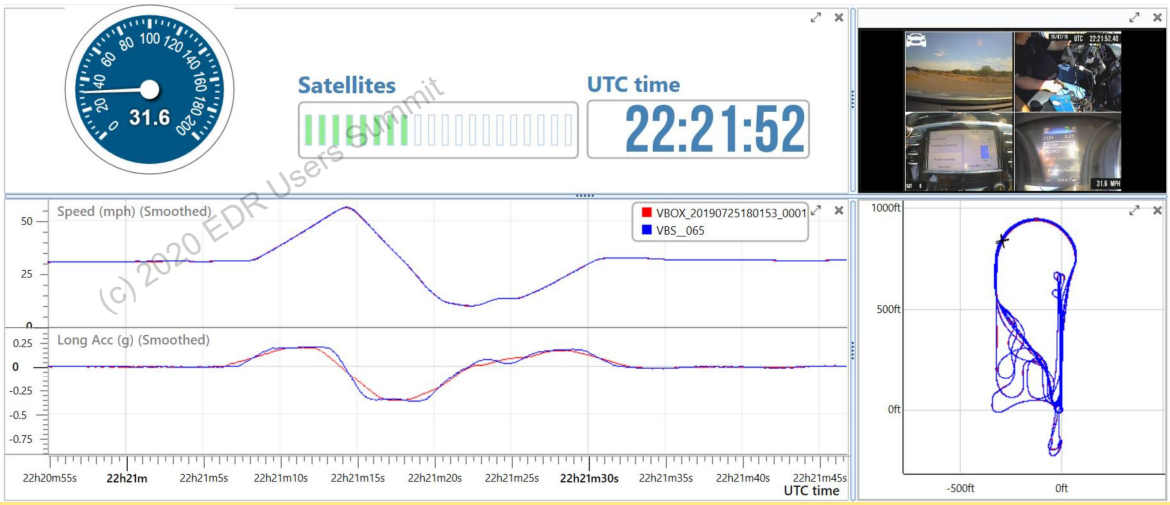


(c) 2020 EDR Users Summit

50

© 2020 – Author and Collision Publishing

Instrumentation: Racelogic VBOX



(c) 2020 EDR Users Summit

51

VCH Triggers

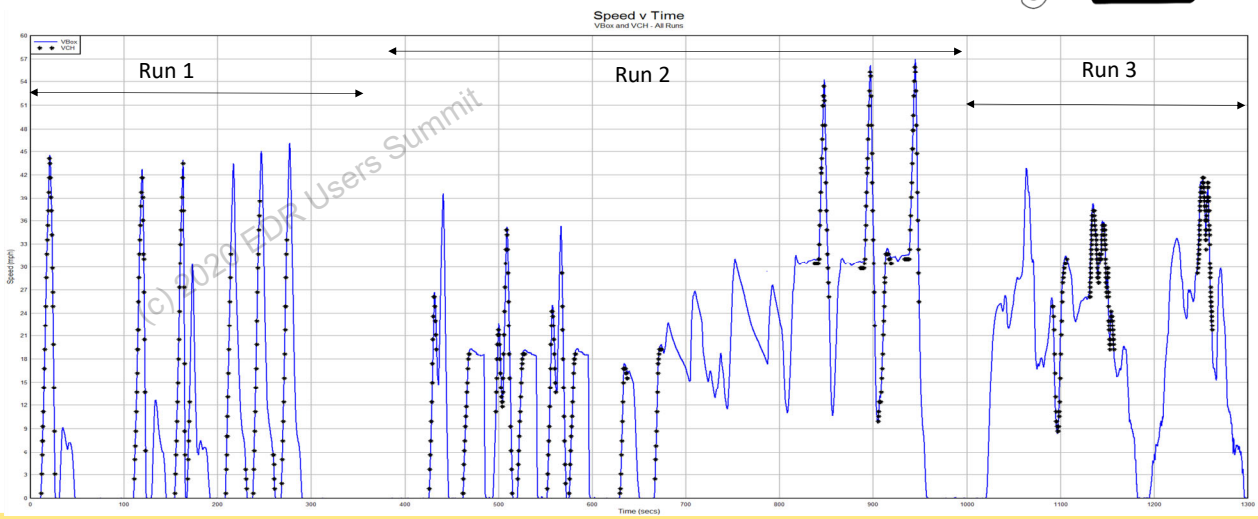
- 2-0, Accelerator high at low speed (4)
- 3-0, Accel high at mid to high speed (5)
- 4-0, Accel high after brake release (13)
- 5-1, Accelerator Med or Higher after shift reverse (3)
- 5-2, Accelerator Med or Higher after shift forward (5)
- 6-2, forward position with med or high accelerator (3)
- 8-0, Simultaneous Accelerator & Brake pedals (3)
- 10-0, medium or high accelerator pedal while in N (1)
- 12-0, VSC (5)
- 13-0, TRC (3)
- 16-0, Sudden turn (4)
- 17-0, Accelerator high with cruise control (3)

(c) 2020 EDR Users Summit

52

© 2020 – Author and Collision Publishing

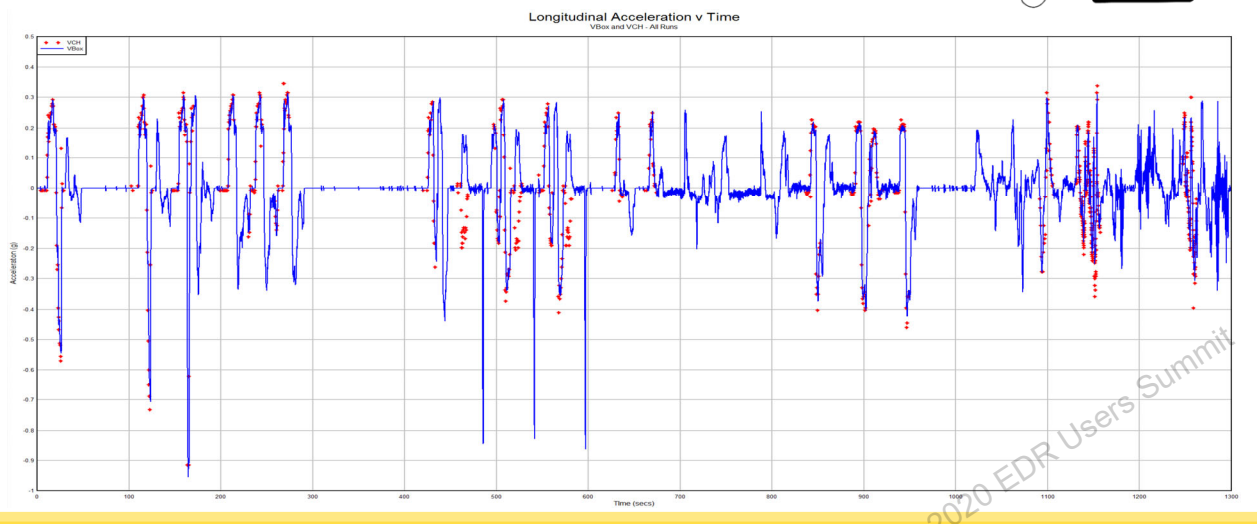
VCH Speed v VBOX



March 9-11, 2020 Slide 53

53

VCH Longitudinal Acceleration v VBOX

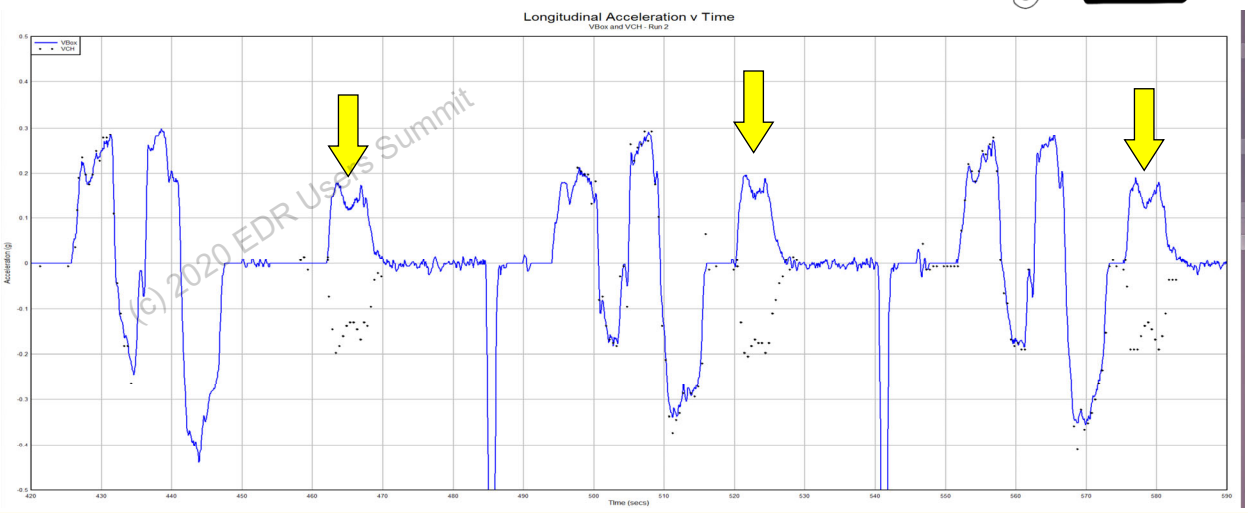


March 9-11, 2020 Slide 54

54

© 2020 – Author and Collision Publishing

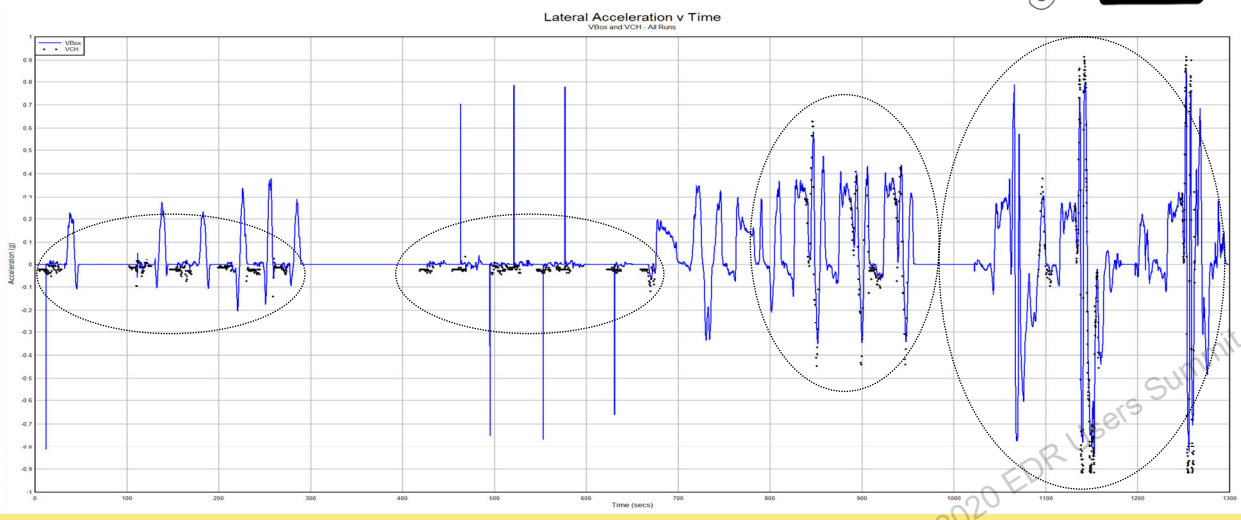
VCH Longitudinal Acceleration v VBOX



March 9-11, 2020 Slide 55

55

VCH Lateral Acceleration v VBOX



March 9-11, 2020 Slide 56

56

© 2020 – Author and Collision Publishing

VCH – How "good" is the data?



- Percent Error:
  - difference from reference measurement/reference measurement
  - Can be misleadingly high when comparing small values
    - e.g., 0.5 is 50% error when the actual reference is 1 mph.

(c) 2020 EDR Users Summit

57

VCH – How "good" is the data?



- R, the correlation coefficient:
  - strength of relationship between variables
  - Values close to 1 are desirable

- RMSE, the Root Mean Square Error:
  - Standard deviation of the errors
  - Values close to 0 are desirable

(c) 2020 EDR Users Summit

58

© 2020 – Author and Collision Publishing

VCH – How "good" is the Speed data?



For all data,
- R = 0.993
- RMSE = 0.69 mph

(c) 2020 EDR Users Summit

Run 1
- R = 0.999
- RMSE = 0.60 mph

Run 2
- R = 0.998
- RMSE = 0.68 mph

Run 3
- R = 0.973
- RMSE = 0.91 mph

59

VCH – How "good" is the accel-x data?



For all data,
- R = 0.806
- RMSE = 0.056 g

Run 1
- R = 0.991
- RMSE = 0.028 g

Run 2
- R = 0.569
- RMSE = 0.088 g

Run 3
- R = 0.967
- RMSE = 0.038 g

(c) 2020 EDR Users Summit

60

© 2020 – Author and Collision Publishing

VCH – How "good" is the accel-y data?

For all data,
- R = 0.506
- RMSE = 0.069 g

Run 1
- R = 0.349
- RMSE = 0.025 g

Run 2
- R = 0.456
- RMSE = 0.036 g

Run 3
- R = 0.931
- RMSE = 0.233 g

(c) 2020 EDR Users Summit

61

Ignition Cycles

Testing and Validation of Toyota Vehicle Control History Data

(c) 2020 EDR Users Summit

62

© 2020 – Author and Collision Publishing

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1249
Multi-Event, Number of Events (times)	0
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

(c) 2020 EDR Users Summit

Techstream

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
01274	00000:37:19	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

63

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1249
Multi-Event, Number of Events (times)	
Time from Event 1 to 2 (sec)	
Diagnostic Trouble Codes Exist	
Location of Side Satellite Sensor 1, L	
Location of Side Satellite Sensor 2, L	
Location of Side Satellite Sensor 3, L	
Location of Side Satellite Sensor 4, L	
Location of Side Satellite Sensor 1, R	
Location of Side Satellite Sensor 2, R	
Location of Side Satellite Sensor 3, R	
Location of Side Satellite Sensor 4, R	
Location of Floor Sensor	Airbag ECU

Current Key Cycle: 01274

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key	
01274	00000:37:19	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

(c) 2020 EDR Users Summit

64

© 2020 – Author and Collision Publishing

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1259
Multi-Event, Number of Events (times)	0
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

(c) 2020 EDR Users Summit

Techstream

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
01284	00000:18:17	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

65

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1259
Multi-Event, Number of Events (times)	
Time from Event 1 to 2 (sec)	
Diagnostic Trouble Codes Exist	
Location of Side Satellite Sensor 1, L	
Location of Side Satellite Sensor 2, L	
Location of Side Satellite Sensor 3, L	
Location of Side Satellite Sensor 4, L	
Location of Side Satellite Sensor 1, R	
Location of Side Satellite Sensor 2, R	
Location of Side Satellite Sensor 3, R	
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

89170-06641
17EDR
Yes
1259

Current Key Cycle
01284

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
01284	00000:18:17	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

(c) 2020 EDR Users Summit

66

© 2020 – Author and Collision Publishing

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1260
Multi-Event, Number of Events (times)	0
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

(c) 2020 EDR Users Summit

Techstream

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
01285	00000:04:05	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

67

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1260
Multi-Event, Number of Events (times)	
Time from Event 1 to 2 (sec)	
Diagnostic Trouble Codes Exist	
Location of Side Satellite Sensor 1, L	
Location of Side Satellite Sensor 2, L	
Location of Side Satellite Sensor 3, L	
Location of Side Satellite Sensor 4, L	
Location of Side Satellite Sensor 1, R	
Location of Side Satellite Sensor 2, R	
Location of Side Satellite Sensor 3, R	
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

Current Key Cycle: 01285

Vehicle Control History

Key Cycle: All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

(c) 2020 EDR Users Summit

68

© 2020 – Author and Collision Publishing

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1263
Multi-Event, Number of Events (times)	0
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

(c) 2020 EDR Users Summit

Techstream

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
01288	00000:06:10	30000

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

69

Ignition Cycles

CDR

System Status at Retrieval	
ECU Part Number	89170-06641
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1263
Multi-Event, Number of Events (times)	
Time from Event 1 to 2 (sec)	
Diagnostic Trouble Codes Exist	
Location of Side Satellite Sensor 1, L	
Location of Side Satellite Sensor 2, L	
Location of Side Satellite Sensor 3, L	
Location of Side Satellite Sensor 4, L	
Location of Side Satellite Sensor 1, R	
Location of Side Satellite Sensor 2, R	
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

Current Key Cycle: 01288

Vehicle Control History

Key Cycle: All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area

(c) 2020 EDR Users Summit

70

© 2020 – Author and Collision Publishing

EDR SUMMIT RESEARCH. COLLECTION. ANALYSIS
MARCH 9-11, 2020

crash·ol·o·gy
THE SCIENCE OF CRASHES

Timestamps

(c) 2020 EDR Users Summit

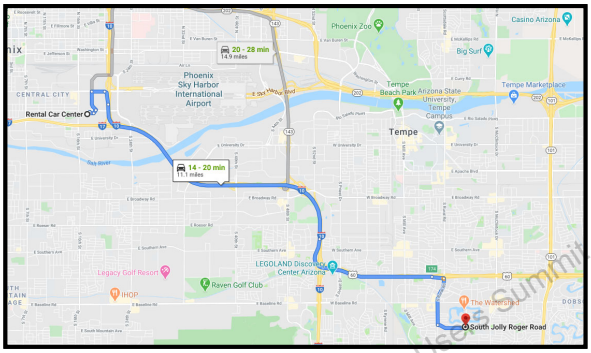
Testing and Validation of Toyota Vehicle Control History Data

71

Timestamps

crash·ol·o·gy
THE SCIENCE OF CRASHES

- 0800 - Vehicle pickup
- 0800-0900 – create events



(c) 2020 EDR Users Summit

72

© 2020 – Author and Collision Publishing

Timestamps



- Approx. 0900 - Vehicle pickup
- Approx. 0930 - create triggers
- Approx. 1015 – baseline acquisition looking for triggers

(c) 2020 EDR Users Summit

73

Timestamps



- Approx. 0900 - Vehicle pickup
- Approx. 0930 - create triggers
- Approx. 1015 – baseline acquisition looking for triggers

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger
01268	00000:11:46.6	-	7/25/2019 11:26:19 AM	Accelerator high position in mid to high speed
01268	00000:12:49.4	-	7/25/2019 11:27:22 AM	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position
01268	00000:12:49.6	-	7/25/2019 11:27:22 AM	Accelerator pedal opening angle signal is high immediately after brake pedal is released
01268	00000:13:19.5	-	7/25/2019 11:27:52 AM	Accelerator high position in mid to high speed
01268	00000:15:39.5	-	7/25/2019 11:30:12 AM	Accelerator signal and brake signal input simultaneously
01268	00000:19:37.3	-	7/25/2019 11:34:10 AM	Accelerator signal and brake signal input simultaneously

(c) 2020 EDR Users Summit

74

© 2020 – Author and Collision Publishing

Timestamps

- Approx. 0900 - Vehicle pickup
- Approx. 0930 - create triggers
- Approx. 1015 – baseline acquisition looking for triggers

Key Cycle	Elapsed Time	Clock Type	Time and Date		Trigger
01268	00000:11:46.6	-	7/25/2019 11:26:19 AM	Accel	
01268	00000:12:49.4	-	7/25/2019 11:27:22 AM	Accel	mediately after shifting to forward position
01268	00000:12:49.6	-	7/25/2019 11:27:22 AM	Accel	tely after brake pedal is released
01268	00000:13:19.5	-	7/25/2019 11:27:52 AM	Accel	
01268	00000:15:39.5	-	7/25/2019 11:30:12 AM	Accel	ly
01268	00000:19:37.3	-	7/25/2019 11:34:10 AM	Accelerator signal and brake signal input simultaneously	

Time and Date
7/25/2019 11:26:19 AM
7/25/2019 11:27:22 AM
7/25/2019 11:27:22 AM
7/25/2019 11:27:52 AM
7/25/2019 11:30:12 AM
7/25/2019 11:34:10 AM

(c) 2020 EDR Users Summit

75

Timestamps



(c) 2020 EDR Users Summit

76

© 2020 – Author and Collision Publishing

Timestamps



(c) 2020 EDR

77

Timestamps

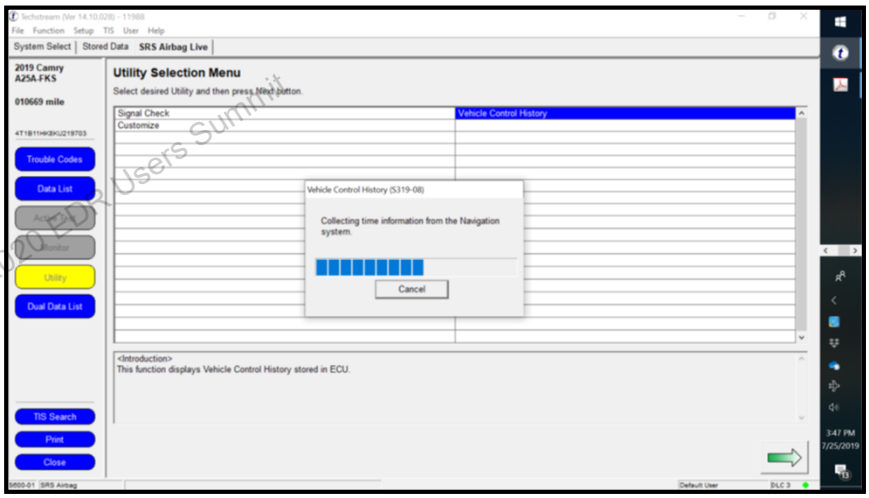


(c) 2020 EDR Users Summit

78

© 2020 – Author and Collision Publishing

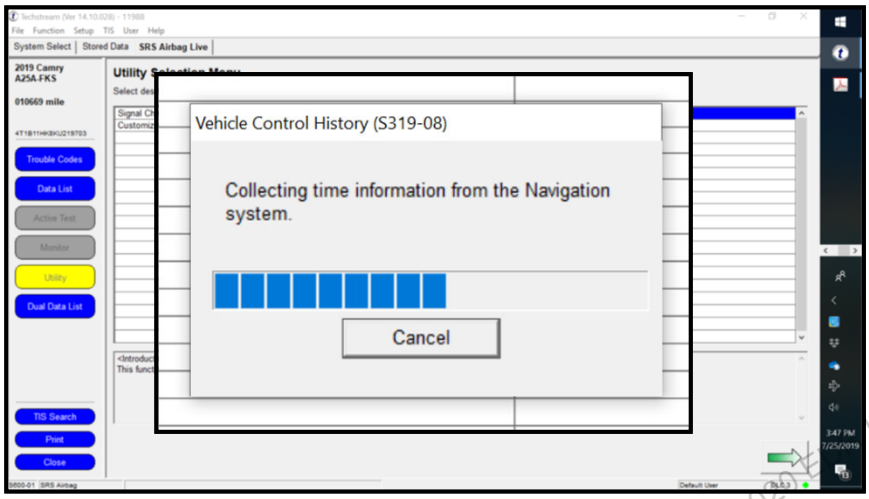
Timestamps



(c) 2020 EDR Users Summit

79

Timestamps



(c) 2020 EDR Users Summit

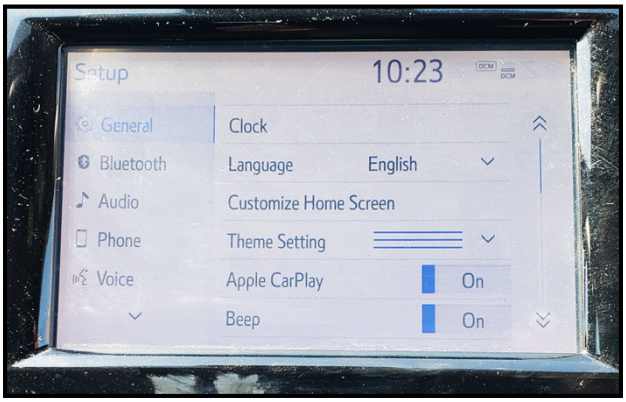
80

© 2020 – Author and Collision Publishing

Timestamps

- Best Practices

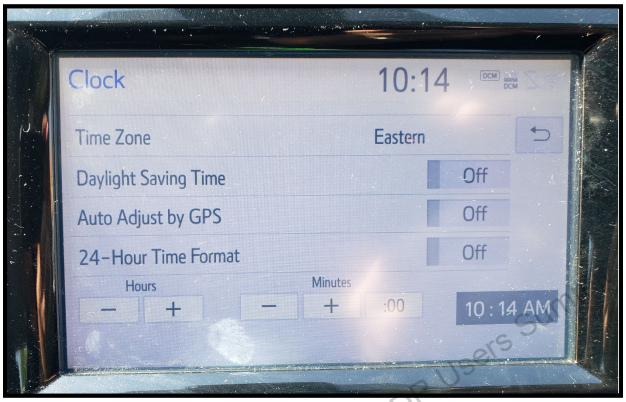
(c) 2020 EDR Users Summit



81

Timestamps

- Best Practices
  - During vehicle inspection document and photograph clock information, including:
    - Time
    - Time Zone
    - DST
    - Auto Adjust Settings



(c) 2020 EDR Users Summit

82